Sweksha Shaw

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EDUCATION

San Jose State University San Jose, CA
Bachelor of Science (B.S.) in Computer Science *Aug 2024 - May 2028*

- *Relevant Coursework:* Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Introduction to AI, Operating Systems, Information Security, Linear Algebra, Calculus I–III

EXPERIENCE

Software Engineering Intern Aug. 2026 - Present
Y STEM and Chess, Inc. | JavaScript, TypeScript, React, Node.js, Express, MongoDB

- Built full-stack chess save/resume for 400 students; the server replays every move to reject faked results, with 3-role access control and per-user rate limits; 300 games saved in the first month
- Averted a prod outage by tracing 100% of API traffic hitting localhost to 39 bad imports & splitting config across 4 services
- Migrated passwords from unsalted SHA-384 to bcrypt by rehashing on login, with 0 forced resets or downtime
- Audited auth across 3 servers, closing 4 vulnerabilities: hardcoded credentials, passwords in URLs, CORS

LLM Research Assistant Aug. 2026 - Present
SJSU College of Science | Python, Ollama, Open-Source LLMs, NVIDIA DGX, GitHub Actions

- Found LLM councils & debate don't beat simple averaging across ~13,000 forecasts from 5 LLMs on 1,039 questions
- Identified self-preference bias through ablations: 4 of 5 chair models over-weighted their own model's forecasts
- Built a fault-tolerant pipeline on an NVIDIA DGX GPU with resumable runs, retries & data audits that caught 5 leaks
- Shipped a public 1,185-question benchmark, a live results dashboard & a one-click Colab reproduction

Full-Stack Website Lead Dec. 2025 - Present
AI & Machine Learning Club, SJSU | TypeScript, Next.js, Node.js, React, PostgreSQL, Supabase

- Built a Postgres-backed analytics dashboard & ML forecasting across 1500+ check-ins, raising average event turnout by 40%
- Cut page load time by 74% (4.6s to 1.2s) by serving avatars via Supabase's image resize API, shrinking payloads by 94%
- Found and patched a privilege escalation in a Postgres trigger letting any of 300+ members become officers
- Automated 5 role-gated CSV exports, including mailing lists and event attendance, cutting data prep time by 70%

PROJECT EXPERIENCE

ToDue Relay | Python, FastAPI, PostgreSQL, SQLAlchemy, AWS (Lambda, EventBridge, SAM), Gemini API, OAuth 2.0

- Built a serverless LLM pipeline on AWS Lambda and EventBridge that extracts action items from Gmail into Google Calendar hourly, auto-scheduling 89% of extractions at 93% precision and routing the rest to human review.
- Cut LLM API calls by 91% with a rules-based pre-filter that screens emails before they reach the model.
- Achieved 0 duplicate events across 500+ runs using atomic PostgreSQL upserts, advisory locks, and idempotent event IDs.
- Automated CI/CD with GitHub Actions and AWS SAM deploys, backed by 340+ integration tests against real PostgreSQL.

Arbiter | TypeScript, Next.js, Node.js, Socket.IO, PostgreSQL, Redis, Docker, Google Cloud

- Launched a real-time group-decision app used by 10+ users in 3 groups; 4-min median decision time, 90% match rate.
- Profiled the server and replaced 31 KB full-state broadcasts with 300 B deltas, doubling capacity to 200 & cutting CPU 36%
- Load-tested real-time voting to 200 concurrent users at 11 ms p95 with zero failures on a single free-tier Google Cloud VM.
- Guaranteed exactly one Google Places API call per session under concurrent requests via an atomic Redis compare-and-set.

TaskMatch (CalHacks 13.0) | Rust, Tauri, Swift, Python, FastAPI, PostgreSQL, Supabase, Llama 3.2, Google Cloud Run

- Shipped a macOS and Windows tray app from one Rust/Tauri codebase that matches users live with someone on the same task
- Built a FastAPI matching API on Supabase Postgres, deployed to Cloud Run, that scored 96% accuracy across 137 test pairs
- Kept raw screen data on-device by running Apple Vision OCR and a local Llama 3.2, uploading only a summary every 5 minutes
- Raised topic-label accuracy from 17% to 92% with an eval harness that runs the app's production Rust summarizer

Dataset Statistical Bias Detector | Python, FastAPI, React, TypeScript, scikit-learn, Docker, GitHub Actions

- Delivered a dataset bias auditor for SJSU's Responsible Computing Club, flagging fairness issues before model training
- Detected hidden bias without labels using PCA, KMeans & IsolationForest, plus χ^2 tests for sensitive-attribute leakage
- Built a modular FastAPI backend with 3 deployable endpoints, Docker & CI running linting, type-checks & pytest

LEADERSHIP EXPERIENCE

Technical Chair @ Society of Women Engineers San Jose, CA

- Collaborated on a points-tracking dashboard, enabling officers to monitor and assess member and event participation.
- Fixed bugs and shipped regular WordPress site updates, helping board and members keep track of events and increasing traffic
- Led professional development workshops, strengthening student readiness for industry.

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, SQL, Java, Rust, Swift, HTML/CSS, Go, C/C++
Tools: PostgreSQL, MongoDB, Redis, Supabase, AWS (Lambda, SAM), Google Cloud (GCP), Docker, GitHub Actions, Ollama
Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Tauri, WordPress
Libraries: Socket.IO, SQLAlchemy, scikit-learn, pytest, Google APIs (Calendar, Places, Gmail, Gemini), REST APIs, OAuth 2.0